Do Now · retrieval

D-type flip-flops

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A half adder receives A = 1 and B = 0. Calculate Sum and Carry. Explain which logic operation produces each output.

Q2

Apply De Morgan’s law to simplify ¬(A ∨ B).

Q3

Insert 40, 20, 60, 30 into an empty binary search tree. State the parent of 30 and whether 30 is its left or right child.

Q4

A = $0.1100_2 \times 2^3$ and B = $0.0100_2 \times 2^3$. Calculate A − B, showing the mantissa calculation and decimal result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Why do we need flip-flops? — Logic gates calculate; a flip-flop remembers. Registers must hold values even when other inputs change.

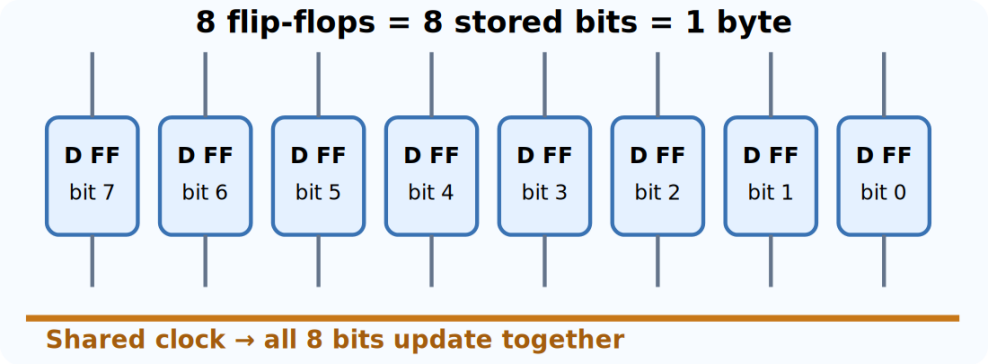
Core idea 2

Supermarket door analogy — D can change before the sample. Q only stores the new value at CLK ↑.

Core idea 3

The rising edge matters — In this lesson we use a rising-edge-triggered D flip-flop: Q copies D only when CLK changes from 0 to 1.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Between clock edges, Q holds — Changing D alone does not change Q. Without a rising edge, the flip-flop keeps its previous bit.

Core idea 5

Trace rising edge 1 — At ↑1, D = 1. Q becomes 1.

Vocabulary · use the correct technical terms

D-type flip-flop — A circuit that stores one bit.

D input — The data value presented to the flip-flop.

clock — A timing signal.

trigger — The clock event that causes sampling.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Model when Q changes — and when it holds

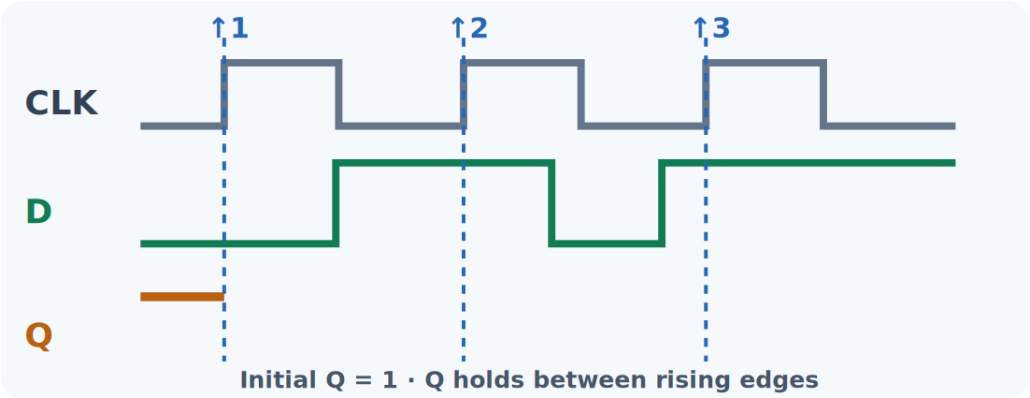
Follow the worked method

1. Identify which edges count — Point to CLK: ↑ is LOW → HIGH and samples D. ↓ is HIGH → LOW and does not update Q. Find all three ↑ edges before reading D.

2. D changes, but Q stays 0 — Look where D rises between ↑1 and ↑2. The clock has not risen again, so Q does not follow D. The falling clock edge also leaves Q unchanged.

3. ↑3: Q stays 1 — despite a rising edge — CLK rises and samples D = 1. But Q already stores 1, so the output does not visibly change. The rule is: at ↑ copy D; otherwise hold Q.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict edge 1
Find D at ↑1. What should Q store next?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

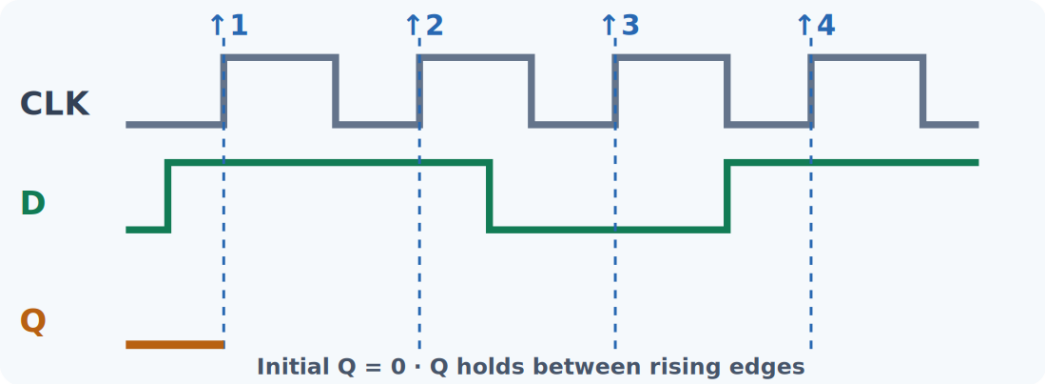
Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

TRACE · 4 MARKS

A rising-edge-triggered D-type flip-flop starts with Q = 0. Trace the different timing diagram and complete Q after each rising edge. [4]

Question 1 · original source diagram



Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

CHALLENGE · 4 MARKS

A 4-bit register initially stores 1010. Just before the next rising edge its D inputs are 0111. Explain the value stored before and after that edge, and what happens if the D inputs subsequently change without another rising edge. [4]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If D changes but no clock trigger occurs, Q should keep its previously stored value.

Use key vocabulary from memory

1. Explain D-type flip-flop in your own words.

2. Explain D input in your own words.

3. Explain clock in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.